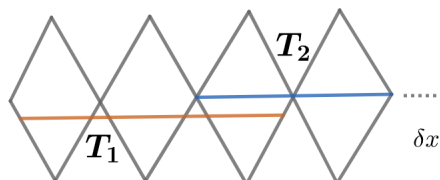


2020B F=ma Exam: Problem 18

Kevin S. Huang



Consider displacing the right end of the arm by δx so that the distance between adjacent intersections increases by $\delta x/4$. By the principle of virtual work, the springs do no work so

$$W_1 + W_2 = -T_1 \left(\frac{3\delta x}{4} \right) - T_2 \left(\frac{2\delta x}{4} \right) = 0$$

Thus,

$$-3T_1 - 2T_2 = 0$$

$$\frac{T_1}{T_2} = -\frac{2}{3}$$

so the answer is $\boxed{A}$.